Lesson Notes

MODULE 5 | LESSON 1

REINTRODUCTION TO GRAPHS AND NETWORKS

Reading Time	30 minutes
Prior Knowledge	Probability, Statistics, Linear Algebra
Keywords	Network Theory, Graph Theory, Bayesian Networks

In the last module, we exploited extreme value theory to enhance our risk assessments. In this lesson, we incorporate graph and network theory to enable more elaborate risk modeling.

1. Where We've Been

So far in this course, we have focused on what might be called traditional elements of risk management: In Module 1, we examined several models for assessing systemic risk, sophisticated models that might nevertheless be considered simplistic given the relatively limited range of factors to which each was sensitive.

In the second module, we examined variance, a concept often used interchangeably with portfolio risk (which may be more appropriate in certain contexts rather than others). We investigated in some detail a popular derivative contract, the variance swap, that can be used to hedge against "market risk" (when this is understood as volatility or variance, as previously mentioned) or speculate on the direction of future market variance or hedge the vega risk in an options book. This study of variance swaps also included a closer look at techniques of replication, another hedging (and therefore risk management practice). In the second half of that module, we learned about and experimented with a type of volatility that jumps and therefore did not fit nicely into our Brownian motion world of Black-Scholes option pricing and mean-variance optimization.

In the third module, we explored two state-of-the-art approaches, applying Deep Learning to Value at Risk modeling. You were already familiar with the concepts of Value at Risk (VaR) and expected shortfall (ES) from previous courses because risk management has always been a part of our portfolio management thinking. Portfolio management always entailed managing the probability of both profit and loss of a portfolio. The "variance" in mean-variance optimization exemplifies how investment growth and risk management are just two sides of the same coin, and that coin is portfolio management.

In the fourth module, we focused on how extreme value theory can improve VaR and ES forecasts. You also already knew the famous GARCH model well from the Econometrics course and then put it to work again in Module 3's Multi-transformer. This allowed us to concentrate on recent research that combined and extended the ideas of VaR, ES, and GARCH, leading to improved versions of these risk measures.

2. Where We are Going: We Have Been Here Before

In this second half of the course, we will be working toward a sophisticated and state-of-the-art approach to risk management that will build on much that you have learned, with special emphasis on probability and statistics (as you might expect) as well as on network theory, upon which we have touched several times in this course and the Portfolio Management course. Also, linear algebra reappears sporadically and insightfully in the rest of the course. For this reason, it will serve us well to review and refine our thinking about probabilistic reasoning and statistical inference and linear algebra. As such, the first parts of this lesson's required readings revisit exactly these foundations.

3. Graphs and Networks

The remainder of this course moves us gradually toward our goal of constructing complex network and scenario analysis, specifically Bayesian networks, to help better manage risk. You will read in this lesson's required reading that a Bayesian network is "a special type of network connecting events that influence each other" (Coscia 29). You can already start to imagine how powerful such a tool might be for a risk manager. You might recall that we have already worked with the concepts of networks and the power of graphs, as recently as Module 1, with its focus on the systemic risk inherent to a financial network. We also engaged productively with graphs in the Portfolio Management course in our search for more robust portfolios. The clustering employed in the lesson "Machine Learning Extensions of Risk Parity" (Portfolio Management, Module 5, Lesson 4) implicitly entailed graph theory, even if we did not create or see any graphic visualizations at that point. In that lesson, we did, however, transform correlations into distances, and what's more, we turned the correlation matrix into a distance matrix, as figure 1 will help you recall.

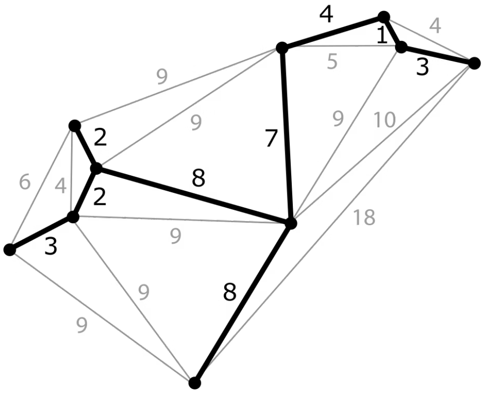
Figure 1: The Distance Matrix

$$A = \begin{bmatrix} 0 & d_{12}^2 & d_{13}^2 & \dots & d_{1n}^2 \\ d_{21}^2 & 0 & d_{23}^2 & \dots & d_{2n}^2 \\ d_{31}^2 & d_{32}^2 & 0 & \dots & d_{3n}^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1}^2 & d_{n2}^2 & d_{n3}^2 & \dots & 0 \end{bmatrix}$$

Source: "Euclidean distance matrix." *Wikipedia*. https://en.wikipedia.org/wiki/Euclidean_distance_matrix.

Later in Portfolio Management—for instance, "Information and Distance," Module 7, Lesson 3—we witnessed the explanatory and predictive power that comes from the merging of information theory and network theory. That same distance matrix is easily turned into a graph, like the simple paradigmatic one in figure 2. The reading for this lesson also discusses how matrices and graphs are basically interoperable or exchangeable (Coscia 68).

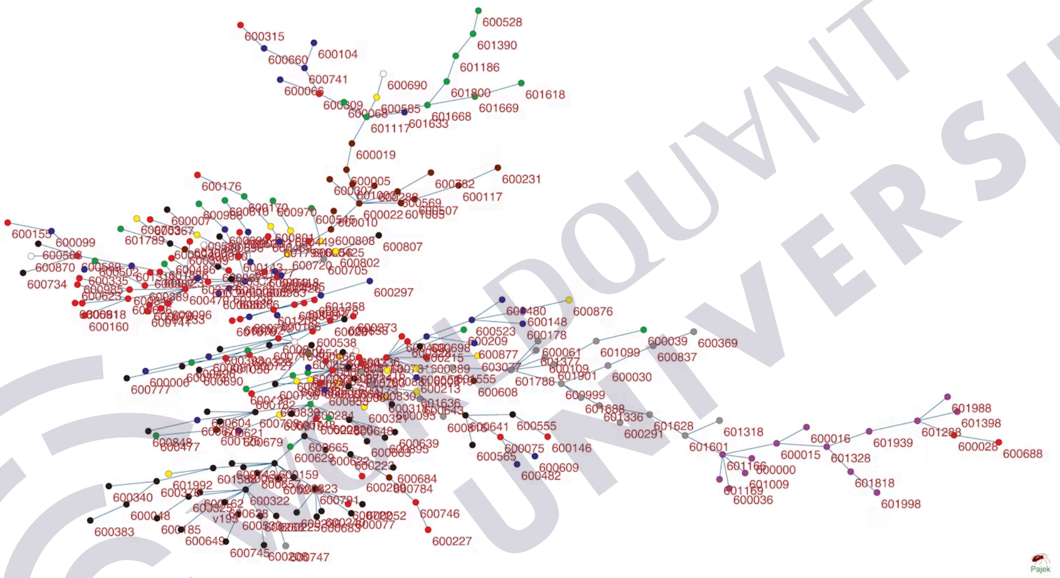
Figure 2: Example of a Minimum Spanning Tree (MST)



Source: "Minimum Spanning Tree." *Wikipedia*, Wikimedia Foundation, 30 Mar 2022. https://en.wikipedia.org/wiki/Minimum_spanning_tree.

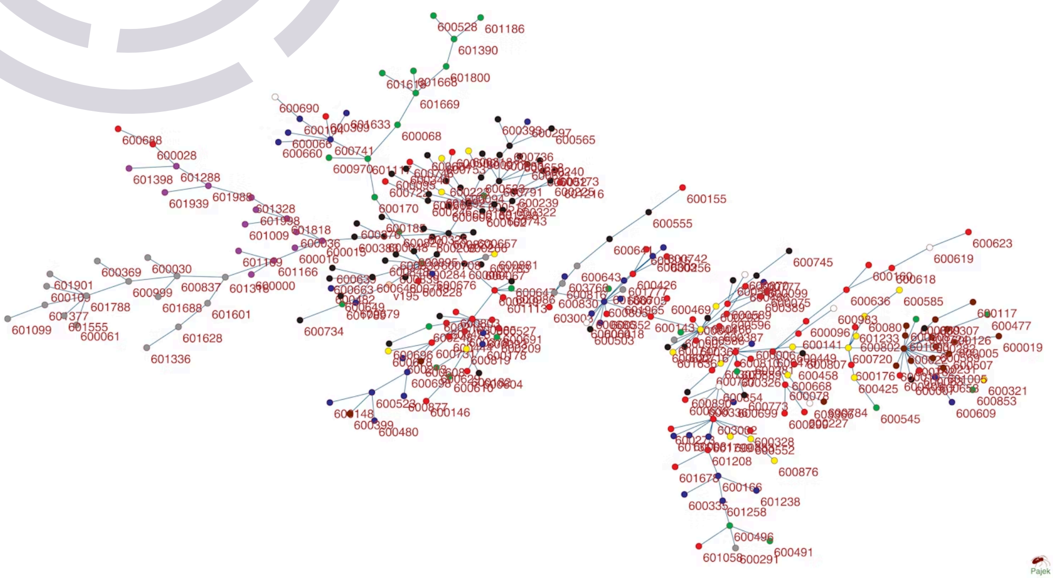
Recall from the required reading for that lesson, with the help of figures 2 and 3, how efficiently the resulting images convey the connectedness of the Shanghai Stock Exchange. Coscia discusses correlation networks specifically in this lesson's required reading (45–46, 64).

Figure 3: Correlation Network Graph



Source: Guo, Xue et al. "Development of Stock Correlation Networks using Mutual Information and Financial Big Data." *PLOS One*, vol. 13, no. 4, 2018, pp. 1–16, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0195941.

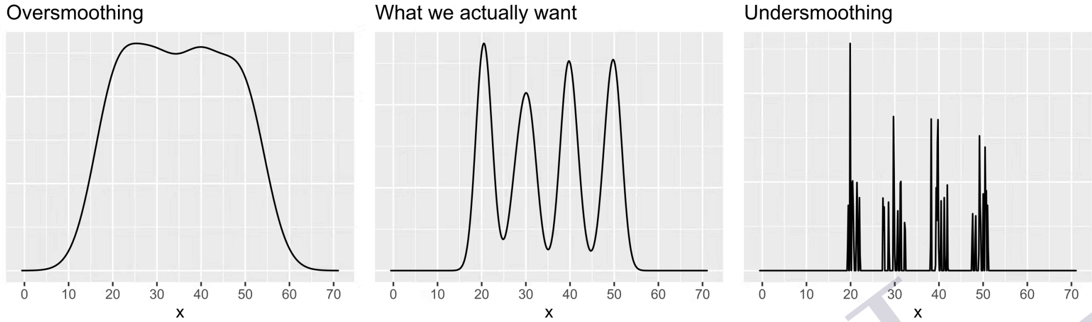
Figure 4: Mutual Information Network Graph



Source: Guo, Xue et al. "Development of Stock Correlation Networks using Mutual Information and Financial Big Data." *PLOS One*, vol. 13, no. 4, 2018, pp. 1–16, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0195941.

Without belaboring the point that we have already seen much of what we will read in this lesson, we offer a few more topics, which are covered in this lesson and were at least touched on before. Of course, information theory references have already made their way into the discussion of graphs and networks above, and that includes mutual information as well as entropy. In Portfolio Management, Module 7, Lesson 1, we also discussed in detail the importance of kernel density estimation, which helps us get to the Goldilocks middle of Figure 5.

Figure 5: The Importance Kernel Density Estimation



Source: Akinshin, Andrey. "The Importance of Kernel Density Estimation Bandwidth." *Aakinshin*, 13 Oct. 2020, https://aakinshin.net/posts/kde-bw/.

4. The Readings

It is debatable whether chapter 1 of Atlas should be included among the required readings for a financial engineering course because it does not involve finance per se—alas, this can be said of much of this particular book. Also, in providing context for what follows, it tends to be more philosophical than you might expect from an engineering required reading. Nevertheless, this chapter emphasizes an important theme from the last several lessons in this Practitioner track: The most powerful concepts and techniques are often actually imports or immigrants from some other field. One example that seems especially apt here is the genetic algorithm discussed in the Portfolio Management course. From the name alone, you can tell that the idea of optimization through mutation and cross-over (or re-combination) comes to us from genetics and evolutionary theory. This idea itself describes the informed, expansive, and stochastic search for better and better tools and models, which defines modern financial engineering as we have attempted to present it in this track. Coscia's introduction makes the point, in so many different ways, that network science is an amalgamation of disparate scientific fields and perspectives, and it is still in the infancy of its own evolution. At least one of the roots of this type of analysis, for example, reaches back to the sociograms, or the mapping of human interpersonal relationships (Coscia 13), alongside the more traditional engineering roots found in the works of Leibniz, Euler, Galilei, Newton, Planck, etc.

You might observe that there is some overlap between the two required readings. It is hoped that you benefit from reading about the same or similar concepts from different (albeit related) perspectives. Given that by now you are already familiar with many of these concepts at least superficially, these are perspectives *in addition to* those to which you have already been exposed. Compare Kochenderfer's Example 2.4 (27) in relation to a discussion of decision trees and Table 2.3 (30) in relation to conditional distributions to Coscia's Figure 2.3 (37) illustrating bits and mutual information. All these concepts are related, but not to be confused or conflated. The point here is just to reiterate the interconnectedness of these apparently somewhat disparate topics. As Coscia says, “everything is a network” (9).

By the way, the relationship (the similarities and the differences) between these ideas would make for a very interesting essay, so please write one!

5. Conclusion

We just started an exciting journey towards expansive risk modeling, reviewing some of our fundamentals and also introducing some basic theory. In the next lesson, we keep exploring this basic theory with the help of Python.

References

- Akinshin, Andrey. "The Importance of Kernel Density Estimation Bandwidth." *Aakinshin*, 13 Oct. 2020, https://aakinshin.net/posts/kde-bw/.
- Coscia, Michele. *The Atlas for the Aspiring Network Scientist*. IT University of Copenhagen, 2021.
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- Kochenderfer, Mykel J. *Algorithms for Decision Making*. MIT Press, 2022.
- "Minimum Spanning Tree." *Wikipedia.* https://en.wikipedia.org/wiki/Minimum_spanning_tree.